

DR. JHONATTAN G. MARTINEZ

Assistant Professor | Digital Construction • Digital Twins • AI/IoT • UAS • Lean Construction • Construction Sustainability

Department of Civil and Architectural Engineering, Aarhus University, Denmark | jgmartinez@cae.au.dk | ORCID: 0000-0001-6715-4440

PROFILE

Industrial engineer and PhD in Engineering Science (Civil Engineering) with 15+ years of research and consultancy experience across Latin America, the United States, Europe, and the Middle East. Research focuses on data-driven construction management and Construction 4.0, integrating Digital Twins, BIM, UAS, IoT sensing, simulation, Lean Construction, AI-enabled decision support, and CO₂/energy monitoring. Experienced in interdisciplinary and industry-collaborative research, research funding, technology demonstrators, postgraduate supervision, and translation of digital methods into construction planning, safety, productivity, and sustainability applications.

ACADEMIC & PROFESSIONAL APPOINTMENTS

- Assistant Professor (Tenure Track), Aarhus University, Denmark | Jan 2024–Present
- Postdoctoral Researcher, Technion – Israel Institute of Technology | Mar 2021–May 2023
- Senior Consultant, PROCASUR Corporation, Chile | Mar 2016–Jun 2020

EDUCATION & RESEARCH TRAINING

Qualification	Period	Institution	Country
Postdoctoral Research Fellow	2021–2023	Technion – Israel Institute of Technology	Israel
PhD, Engineering Sciences (Civil Engineering)	2016–2020	Pontifical Catholic University of Chile	Chile
Research Scholar, Construction Management	2018–2020	University of Florida	United States
MBA	2012–2013	National University	Colombia
Industrial Engineering	2002–2007	University of the Atlantic	Colombia

RESEARCH EXPERTISE

Digital Twins • Construction 4.0 • AI-enabled decision support • BIM • UAS/UAV remote sensing • IoT sensing • simulation and agent-based modelling • Lean Construction • production planning and control • construction safety • productivity • energy and CO₂ monitoring • sustainable construction

SELECTED RESEARCH FUNDING & PROJECT LEADERSHIP

- **Smart Infrastructure Emissions Intelligence (SIEI): A Decision-Support System for Sustainable Civil Infrastructure** Principal Investigator, Danish Business Authority (2026–2028)
- **GIS-based Platform Utilized for Seismic Resilience of Urban Communities (GURU)** Co-PI, EUREKA (2025/26–2028)
- **NextGen Build: Empowering Green Innovation and Carbon Intelligence in African and Danish Cities** — Principal Investigator, Danida (2026–2027)
- **Global Innovation Network Program (GINP)** Co-Investigator. Danish Agency for Higher Education and Science (2025–2026)
- **Predictive Carbon Emissions Monitoring and Control System for Earthwork and Interior Work** — Principal Investigator, We Build Denmark (2024–2025)
- **BIM2TWIN: Optimal Construction Management & Production Control** Senior Researcher, Horizon Europe (2020–2023)
- **Young Leaders Program for Rural Development in Central America** Senior Consultant, IFAD (2019–2020)
- **Advanced Monitoring of Construction Safety Using UAS and 3D Models**, Principal Investigator, Pontifical Catholic University of Chile (2018)
- **Strengthening Capacities and Tools to Disseminate Innovations**, Senior Consultant, IFAD (2016–2018)

SELECTED JOURNAL PUBLICATIONS

- Martinez, J. G., Murguía, D., Osorio, C., Gimenez, Z., & Yeung, T. Digital twin construction readiness: Development and validation of an organizational assessment instrument. *Journal of Construction Engineering and Management*. Under review.
- Loyola, L. I., & Martinez, J. G. (2025). Real-time energy and CO₂ optimization in construction cranes through digital twins and lean construction. *Revista Ingeniería de Construcción*, 40(Special Issue), 1–11. <https://doi.org/10.7764/RIC.00166.21>
- Yeung, T., Martinez, J. G., Schlenger, J., Borrmann, A., & Sacks, R. (2025). Integrating digital twin and agent-based simulation to support adaptive system design in building projects. *Automation in Construction*, 180, 106550. <https://doi.org/10.1016/j.autcon.2025.106550>
- Martinez, J., Yeung, T., Sacks, R., Shahaf, Y., & Sharoni, L. (2023). Situational awareness in construction uses a serious game. *Journal of Construction Engineering and Management*. <https://doi.org/10.1061/JCEMD4.COENG-12521>
- Martinez, J., Albeaino, G., Gheisari, M., Issa, R. R. A., & Alarcón, L. F. (2021). iSafeUAS: An unmanned aerial system for construction safety inspection. *Automation in Construction*, 125, 103595. <https://doi.org/10.1016/j.autcon.2021.103595>
- Martinez, J., Albeaino, G., Gheisari, M., Volkmann, W., & Alarcón, L. F. (2021). UAS point cloud accuracy assessment using SfM-based photogrammetry and PPK georeferencing technique for building surveying applications. *Journal of Computing in Civil Engineering*. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000936](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000936)
- Martinez, J., Gheisari, M., & Alarcón, L. F. (2020). UAV integration in current construction safety planning and monitoring processes. *Journal of Management in Engineering*, 36(3), 05020005. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000761](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000761)

BOOK / BOOK CHAPTER

- Martinez, J. G., Alarcón, L. F., & Wandahl, S. (2024). Unmanned Aerial Systems (UAS)-Derived 3D Models for Digital Twin Construction Applications. <https://doi.org/10.5772/intechopen.1004746>

RESEARCH IMPACT & ACADEMIC INDICATORS

- 7 journal publications with selective editorial policy; 15 conference publications; 2 book chapters; 1 book.
- 590+ Google Scholar citations (reported September 2026).
- Master's supervision: 10 completed dissertations and 3 ongoing.
- Technology translation: 2 software prototypes for construction planning, optimization, and CO₂ emissions; 3 optimized processes implemented.
- Industry engagement: technical/scientific consultancy with 22 organizations across three continents; 2 companies created/supported.

TEACHING, SUPERVISION & SERVICE

- Teaching: Construction 4.0, Aarhus University; topics include BIM/CDE, cyber-physical systems, Digital Twins, IoT, autonomous construction, robotics, and extended reality.
- Postgraduate supervision in digital construction, resilience, construction safety, sensing, Digital Twins, productivity, and sustainability.
- Reviewer for *Automation in Construction*, *Advanced Engineering Informatics*, and *Journal of Management in Engineering*.
- International research collaborations and networks spanning Europe, North America, Latin America, and the Middle East.

SELECTED AWARDS

- Best UAS design for safety applications, IDC 2020.
- Postdoctoral and doctoral scholarships supporting international research training.

ONLINE PROFILES

Aarhus University profile: [https://cae.au.dk/en/about-the-](https://cae.au.dk/en/about-the-department/contact/employees/show/person/jgmartinez%40cae.au.dk)

[department/contact/employees/show/person/jgmartinez%40cae.au.dk](https://cae.au.dk/en/about-the-department/contact/employees/show/person/jgmartinez%40cae.au.dk) | **ORCID:** <https://orcid.org/0000-0001-6715-4440> |

Google Scholar: <https://scholar.google.com/citations?user=iSNe3tQAAAAJ&hl=en>